EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then -

(A) $A = B$ (B) $A \subset B$ and $B - A \neq \phi$

(C) $A \not\subset B$ (D) $B \not\subset A$

MTE003

2. If $\frac{1-\cos 2\theta}{1+\cos 2\theta} = 3$, then the general solution of θ is –

(A) $2n\pi \pm \frac{\pi}{6}$ (B) $n\pi \pm \frac{\pi}{6}$

(C) $2n\pi \pm \frac{\pi}{3}$ (D) $n\pi \pm \frac{\pi}{3}$

where $n \in I$

MTE002

3. The solution set of $(5 + 4\cos\theta)(2\cos\theta + 1) = 0$ in the interval $[0,2\pi]$ is :

(A) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$ (B) $\left\{\frac{\pi}{3}, \pi\right\}$

(C) $\left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$ (D) $\left\{\frac{2\pi}{3}, \frac{5\pi}{3}\right\}$

MTE004

4. The number of solutions of the equation $\sin 2x - 2\cos x + 4\sin x = 4$ in the interval $[0, 5\pi]$ is -

(A) 6 (B) 4 (C) 3 (D) 5

MTE007

5. If $2\tan^2\theta = \sec^2\theta$, then the general solution of θ -

(A) $n\pi + \frac{\pi}{4}\ (n \in I)$ (B) $n\pi - \frac{\pi}{4}\ (n \in I)$

(C) $n\pi \pm \frac{\pi}{4}\ (n \in I)$ (D) $2n\pi \pm \frac{\pi}{4}\ (n \in I)$

MTE001

6. The general solution of equation $4\cos^2 x + 6\sin^2 x = 5$ is -

(A) $x = n\pi \pm \frac{\pi}{2}\ (n \in I)$ (B) $x = n\pi \pm \frac{\pi}{4}\ (n \in I)$

(C) $x = n\pi \pm \frac{3\pi}{2}\ (n \in I)$ (D) None of these

MTE005

7. If $\tan^2\theta - (1 + \sqrt{3})\tan\theta + \sqrt{3} = 0$, then the general value of θ is :

(A) $n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{3}$ (B) $n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{3}$

(C) $n\pi + \frac{\pi}{4}, n\pi - \frac{\pi}{3}$ (D) $n\pi - \frac{\pi}{4}, n\pi - \frac{\pi}{3}$

where $n \in I$

MTE006

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8. The set of angles between 0 and 2π satisfying the equation $4\cos^2\theta - 2\sqrt{2}\cos\theta - 1 = 0$ is

(A) $\left\{\frac{\pi}{12}, \frac{5\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$ (B) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}\right\}$

(C) $\left\{\frac{5\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}\right\}$ (D) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$

MTE008

9. If $\frac{\tan 2\theta + \tan\theta}{1 - \tan\theta\tan 2\theta} = 0$, then the general value of θ is -

(A) $n\pi; n \in I$ (B) $\frac{n\pi}{3}; n \in I$

(C) $\frac{n\pi}{4}$ (D) $\frac{n\pi}{6}; n \in I$

where $n \in I$

MTE010

10. If $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta\tan 4\theta\tan 7\theta$, then θ =

(A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$

(C) $\frac{n\pi}{12}, n \neq (12m-6), n, m \in I$ (D) $n\pi$

where $n \in I$

MTE009

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11. If $\tan\theta - \sqrt{2}\sec\theta = \sqrt{3}$, then the general solution of θ is -

(A) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$ (B) $n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$

(C) $n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{4}$ (D) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$

where $n \in I$

MTE011

12. The most general values of x for which $\sin x + \cos x = \min_{a \in R} \{1, a^2 - 4a + 6\}$ is given by -

(A) $2n\pi$ (B) $2n\pi + \frac{\pi}{2}$

(C) $n\pi + (-1)^n \cdot \frac{\pi}{4} - \frac{\pi}{4}$ (D) None of these

where $n \in I$

MTE012

13. Number of values of x satisfying the equation $\log_2(\sin x) + \log_{1/2}(-\cos x) = 0$ in the interval $(-\pi, \pi]$ is equal to -

(A) 0 (B) 1 (C) 2 (D) 3

MTE014

14. If the equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$ has a solution then k must lie in the interval:

(A) $(-4, -2)$ (B) $[-3, 2)$ (C) $(-4, -3)$ (D) $[-3, -2]$

MTE013

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15. If $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$, then the greatest positive solution of $1 + \sin^4 x = \cos^2 3x$ is -

(A) π (B) 2π (C) $\frac{5\pi}{2}$ (D) none of these

MTE015

16. The number of solutions of the equation $\sin x = x^2 + x + 1$ is-

(A) 0 (B) 1 (C) 2 (D) None

MTE016

17. The number of solutions of the equation $2\cos\left(\frac{x}{2}\right) = 3^x + 3^{-x}$ is -

(A) 1 (B) 2 (C) 3 (D) None

MTE017

18. The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}, 0 < x \le \frac{\pi}{2}$ has

(A) one real solutions (B) more than one real solutions
(C) no real solution (D) none of the above

MTE018

19. Given $a^2 + 2a + \text{cosec}^2\left(\frac{\pi}{2}(a+x)\right) = 0$ then, which of the following holds good ?

(A) $a = 1; \frac{x}{2} \in I$ (B) $a = -1; \frac{x}{2} \in I$
(C) $a \in R\ ; x \in \phi$ (D) a, x are finite but not possible to find

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MTE019

20. The complete solution set of the inequality $\tan^2 x - 2\sqrt{2}\tan x + 1 \le 0$ is -

(A) $n\pi + \frac{\pi}{8} \le x \le \frac{3\pi}{8} + n\pi,\ n \in I$ (B) $n\pi + \frac{\pi}{4} \le x \le \frac{3\pi}{4} + n\pi,\ n \in I$
(C) $n\pi + \frac{\pi}{16} \le x \le \frac{3\pi}{8} + n\pi,\ n \in I$ (D) $n\pi + \frac{\pi}{3} \le x \le \frac{2\pi}{3} + n\pi,\ n \in I$

MTE020

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The equation $2\sin\frac{x}{2}.\cos^2 x + \sin^2 x = 2\sin\frac{x}{2}.\sin^2 x + \cos^2 x$ has a root for which

(A) $\sin 2x = 1$ (B) $\sin 2x = -1$

(C) $\cos x = \frac{1}{2}$ (D) $\cos 2x = -\frac{1}{2}$

MTE041

2. Which of the following set of values of x satisfies the equation $2^{(2\sin^2 x - 3\sin x + 1)} + 2^{(2 - 2\sin^2 x + 3\sin x)} = 9$ is -

(A) $x = x\pi, n \in I$ (B) $x = 2n\pi + \frac{\pi}{2}, n \in I$

(C) $x = m\pi \pm \frac{\pi}{6}, n \in I$ (D) $x = n\pi \pm \frac{\pi}{3}, n \in I$

MTE042

3. If sum of all the solution of the equation $\cot x + \operatorname{cosec} x + \sec x = \tan x$ in $[0, 2\pi]$ is $\frac{k\pi}{2}$, then the value of k is greater than

(A) 1 (B) 2 (C) 3 (D) 4

MTE043

4. $\cos 4x \cos 8x - \cos 5x \cos 9x = 0$ if

(A) $\cos 12x = \cos 14x$ (B) $\sin 13x = 0$

(C) $\sin x = 0$ (D) $\cos x = 0$

MTE048

5. The equation $\sin x + \cos(k + x) + \cos(k - x) = 2$ has real solution(s), then $\sin k$ can be

(A) $\frac{-3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

MTE044

6. The equation $\sin^6 x + \cos^6 x = a^2$ has real solution if

(A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$

(C) $a \in \left(-1, \frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$

MTE047

7. If $m > 0, n < 5, 0 < m + n < 10$ and $\frac{x^2 + 5}{2} = x - 2\cos(m + nx)$ has atleast one real root, then

(A) the greatest value of $(m + n)$ is 3π.
(B) the greatest value of $(m + n)$ is 2π.
(C) the least value of $(m + n)$ is π.
(D) the least value of $(m + n)$ is 2π.

MTE045

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8. If $x^4 + 3\cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$ then

(A) $a + b + c = \pi$ (B) $a - b + c = \pi$

(C) $a + b + c = 3\pi$ (D) $a - b + c = 3\pi$

MTE046

9. If the equation $\cot^4 x - 2\text{cosec}^2 x + a^2 = 0$ has atleast one solution then, sum of all possible integral values of 'a' is not equal to

(A) 1 (B) 0

(C) 3 (D) 2

MTE126

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 10 & 11

Let $f(x) = \cos x + \sin x - 1$ and $g(x) = \sin 2x - 2$.

10. Number of solutions of the equation $f(x) = g(x)$ in $x \in \left[-\pi, \frac{7\pi}{2}\right]$ is equal to

(A) 4 (B) 5

(C) 6 (D) 8

MTE049

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11. If the equation $f(x) = k$ has atleast one real solution then the number of possible integral values of k is equal to

(A) 1 (B) 2

(C) 3 (D) 4

MTE050

Paragraph for Question No. 12 & 13

Let $f(x) = \sin^2 x - (a - 1)\sin x + 2(a - 3)$

On the basis of above information, answer the following questions :

12. If $x \in [0, \pi]$ and $f(x) = 0$ has exactly one real root, then 'a' lies in

(A) (3, 5) (B) (2, 4)

(C) (4, 5) (D) none of these

MTE051

13. If $f(x) = 0$ have two real roots in $(0, \pi)$, then $a \in$

(A) (1, 2) (B) (3, 4)

(C) $(3, 4) \cup \{5\}$ (D) (3, 5)

MTE052

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MATCHING LIST TYPE QUESTION

14.

	List-I		**List-II**
(I)	Number of common solutions of the equations $2\cos^2 x - 3\cos x + 1 = 0$ and $\tan\left(\frac{3x}{4}\right)+1 = 0$, where $-\pi < x \le 3\pi$, is equal to	(P)	2
(II)	Number of solutions of the equation $\sin^2 x + 2\sin x - 4\cos x - \sin 2x = 0$ in $(0, 2\pi)$ is equal to	(Q)	12
(III)	If the sum of all possible values of $x \in (0, 2\pi)$ satisfying the equation $2\cos x \operatorname{cosec} x - 4\cos x - \operatorname{cosec} x = -2$ is equal to $\frac{k\pi}{4}$ $(k \in N)$, then the value of k is	(R)	4
		(S)	3
(IV)	Let $x - y = \frac{1}{4}$ and $\cos(\pi x)\cdot\cos(\pi y) = \frac{1}{2\sqrt{2}}$ where $x, y \in (0, 2)$ then the number of ordered pair(s) (x, y) is equal to	(T)	1

(A) I → P; II → P; III → Q; IV → R
(B) I → T; II → P; III → S; IV → R
(C) I → R; II → P; III → S; IV → T
(D) I → Q; II → P; III → S; IV → T

MTE053

MATRIX MATCH TYPE QUESTION

15.

	Column I		**Column II**
(A)	The number of real roots of the equation $\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is	(P)	1

MTE054

(B)	The number of solutions of the equation $\lvert\cot x\rvert = \cot x + \frac{1}{\sin x}$ $(0 < x < \pi)$ is	(Q)	2
		(R)	0

MTE055

(C)	The number of solution(s) of equation in $(0, 3\pi)$ $\sin^2 x + 3\sec^2 x = -(\tan^2 x + \cot^2 x)$ is	(S)	3

MTE056

(D)	The number of values of $x \in [-2\pi, 2\pi]$, which satisfy $\operatorname{cosec} x = 1 + \cot x$	(T)	5

MTE057

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EXERCISE (S)

1. Number of principal solution(s) of the equation $4 \cdot 16^{\sin^2 x} = 2^{6\sin x}$ is

MTE021

2. The number of solutions of the equation $\tan^2 x - \sec^{10} x + 1 = 0$ in $(0, 10)$ is -

MTE022

3. If α, β $(\alpha - \beta \neq 2n\pi, n \in I)$ are different values of θ satisfying the equation $5\cos\theta - 12\sin\theta = 11$.

If the value of $\sin(\alpha + \beta) = -\frac{5k}{169}$, then the value of k is

MTE059

4. Find the number of solutions of the equation $\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$ in the interval $[0, 2\pi]$.

MTE071

5. In $(0, 6\pi)$, find the number of solutions of the equation $\tan\theta + \tan 2\theta + \tan 3\theta = \tan\theta.\tan 2\theta.\tan 3\theta$

MTE070

6. If $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $4\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right) + \sqrt{4\sin^4 x + \sin^2 2x}$ is always equal to

MTE060

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7. If $\sin \pi x + \cos \pi x = 0$, then the sum of all solutions in $[0, 100]$ is

MTE058

8. If the arithmetic mean of the roots of the equation $4\cos^3 x - 4\cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval $[0, 315]$ is equal to $k\pi$, then find the value of k

MTE061

9. If $0 \le x \le 3\pi$, $0 \le y \le 3\pi$ and $\cos x \cdot \sin y = 1$, then the possible number of values of the ordered pair (x, y) is -

MTE023

10. Number of integral solution(s) of the inequality $2\sin^2 x - 5\sin x + 2 > 0$ in $x \in [0, 2\pi]$, is -

MTE025

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EXERCISE - JEE (Main) PYQ

1. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are: **[AIEEE 2011]**

(1) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$ (2) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(3) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$ (4) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

MTE026

2. In a ΔPQR, if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to : **[AIEEE 2012]**

(1) $\frac{3\pi}{4}$ (2) $\frac{5\pi}{6}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{4}$

MTE027

3. If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is :- **[JEE (Main) 2016]**

(1) 9 (2) 3 (3) 5 (4) 7

MTE028

4. If sum of all the solutions of the equation $8\cos x \cdot \left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)=1$ in $[0, \pi]$ is $k\pi$, then k is equal to : **[JEE (Main) 2018]**

(1) $\frac{13}{9}$ (2) $\frac{8}{9}$ (3) $\frac{20}{9}$ (4) $\frac{2}{3}$

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MTE029

5. The number of solutions of the equation $1+\sin^4 x=\cos^2 3x,\ x\in\left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is : **[JEE (Main) 2019]**

(1) 5 (2) 4 (3) 7 (4) 3

MTE030

6. Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to : **[JEE (Main) 2019]**

(1) [2, 6] (2) [3,7] (3) R (4) [1,4]

MTE031

7. If $\sin^4\alpha+4\cos^4\beta+2=4\sqrt{2}\sin\alpha\cos\beta;\ \alpha,\beta\in[0,\pi]$, then $\cos(\alpha+\beta)-\cos(\alpha-\beta)$ is equal to : **[JEE (Main) 2019]**

(1) 0 (2) $-\sqrt{2}$ (3) -1 (4) $\sqrt{2}$

MTE032

8. If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval : **[JEE (Main) 2020]**

(1) $\left[-\frac{3}{2}, -\frac{5}{4}\right]$ (2) $\left(-\frac{1}{2}, -\frac{1}{4}\right]$ (3) $\left(-\frac{5}{4}, -1\right)$ (4) $\left[-1, -\frac{1}{2}\right]$

MTE033

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9. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in : **[JEE (Main) 2021]**

(1) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ (2) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(3) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$ (4) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

MTE034

10. If n is the number of solutions of the equation $2\cos x\left(4\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)-1\right)=1, x \in [0, \pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is : **[JEE (Main) 2021]**

(1) $(3, 13\pi / 9)$ (2) $(2, 2\pi / 3)$ (3) $(2, 8\pi / 9)$ (4) $(3, 5\pi / 3)$

MTE035

11. The number of roots of the equation, $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to : **[JEE (Main) 2021]**

(1) 3 (2) 4 (3) 8 (4) 2

MTE036

12. The number of solutions of the equation $\cos\left(x+\frac{\pi}{3}\right)\cos\left(\frac{\pi}{3}-x\right)=\frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is : **[JEE (Main) 2022]**

(1) 8 (2) 5 (3) 6 (4) 7

MTE037

13. Let $S=\left\{\theta\in[-\pi,\pi]-\left\{\pm\frac{\pi}{2}\right\}:\sin\theta\tan\theta+\tan\theta=\sin 2\theta\right\}$. If $T=\sum_{\theta\in S}\cos 2\theta$, then $T + n(S)$ is equal **[JEE (Main) 2022]**

(1) $7+\sqrt{3}$ (2) 9 (3) $8+\sqrt{3}$ (4) 10

MTE038

14. Let $S=\left\{\theta\in\left(0,\frac{\pi}{2}\right):\sum_{m=1}^{9}\sec\left(\theta+(m-1)\frac{\pi}{6}\right)\sec\left(\theta+\frac{m\pi}{6}\right)=-\frac{8}{\sqrt{3}}\right\}$ Then **[JEE (Main) 2022]**

(1) $S=\left\{\frac{\pi}{12}\right\}$ (2) $S=\left\{\frac{2\pi}{3}\right\}$ (3) $\sum_{\theta\in S}\theta=\frac{\pi}{2}$ (4) $\sum_{\theta\in S}\theta=\frac{3\pi}{4}$

MTE039

15. The number of elements in the set $S=\left\{x\in\mathbb{R}:2\cos\left(\frac{x^2+x}{6}\right)=4^x+4^{-x}\right\}$ is: **[JEE (Main) 2022]**

(1) 1 (2) 3 (3) 0 (4) infinite

MTE040

16. The number of elements in the set $S = \{\theta \in [0,2\pi]: 3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0\}$ is **[JEE (Main) 2023]**

(1) 10 (2) 8 (3) 9 (4) 12

MTE073

17. If m and n respectively are the numbers of positive and negative values of θ in the interval $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to. **[JEE (Main) 2023]**

MTE074

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18. If the solution of the equation $\log_{\cos x}\cot x+4\log_{\sin x}\tan x=1, x\in\left(0,\frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers and $\sqrt{\beta}\notin Q$ then $\alpha+\beta$ is equal to: **[JEE (Main) 2023]**

(1) 3 (2) 5 (3) 6 (4) 4

MTE075

19. Let $S=\left\{x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right): 9^{1-\tan^2 x}+9^{\tan^2 x}=10\right\}$ and $\beta=\sum_{x\in s}\tan^2\left(\frac{x}{3}\right)$, then $\frac{1}{6}(\beta-14)^2$ is equal to **[JEE (Main) 2023]**

(1) 32 (2) 8 (3) 64 (4) 16

MTE076

20. Let $S = \{\theta \in [0, 2\pi) : \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$. Then $\sum_{\theta\in S}\sin^2\left(\theta+\frac{\pi}{4}\right)$ is equal to **[JEE (Main) 2023]**

MTE077

21. The number of solutions of $\sin^2 x + (2 + 2x - x^2)\sin x - 3(x - 1)^2 = 0$, where $-\pi \le x \le \pi$, is **[JEE (Main) 2024]**

MTE127

22. Let $\left|\cos\theta\cos(60-\theta)\cos(60-\theta)\right|\le\frac{1}{8}, \theta\in[0,2\pi]$

Then, the sum of all $\theta\in[0,2\pi]$, where $\cos 3\theta$ attains its maximum value, is : **[JEE (Main) 2024]**

(1) 9π (2) 18π (3) 6π (4) 15π

MTE128

23. The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ is : **[JEE (Main) 2024]**

(1) 1 (2) 3 (3) 2 (4) 0

MTE129

24. If $2\tan^2\theta-5\sec\theta=1$ has exactly 7 solutions in the interval $\left[0,\frac{n\pi}{2}\right]$, for the least value of $n\in N$ then $\sum_{k=1}^{n}\frac{k}{2^k}$ is equal to : **[JEE (Main) 2024]**

(1) $\frac{1}{2^{15}}(2^{14}-14)$ (2) $\frac{1}{2^{14}}(2^{15}-15)$ (3) $1-\frac{15}{2^{13}}$ (4) $\frac{1}{2^{13}}(2^{14}-15)$

MTE130

25. If $2\sin^3 x + \sin 2x\cos x + 4\sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0,\frac{n\pi}{2}\right]$, $n\in N$, then the roots of the equation $x^2+nx+(n-3)=0$ belong to : **[JEE (Main) 2024]**

(1) $(0,\infty)$ (2) $(-\infty,0)$ (3) $\left(-\frac{\sqrt{17}}{2},\frac{\sqrt{17}}{2}\right)$ (4) Z

MTE131

26. If $\alpha, -\frac{\pi}{2}<\alpha<\frac{\pi}{2}$ is the solution of $4\cos\theta+5\sin\theta=1$, then the value of $\tan\alpha$ is **[JEE (Main) 2024]**

(1) $\frac{10-\sqrt{10}}{6}$ (2) $\frac{10-\sqrt{10}}{12}$ (3) $\frac{\sqrt{10}-10}{12}$ (4) $\frac{\sqrt{10}-10}{6}$

MTE132

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EXERCISE - JEE (Advanced) PYQ

1. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has **[JEE (Advanced) 2014]**

(A) infinitely many solutions (B) three solutions

(C) one solution (D) no solution

MTE062

2. The number of distinct solutions of equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is **[JEE (Advanced) 2015]**

MTE063

3. Let $S = \left\{x \in (-\pi, \pi) : x \neq 0, \pm\frac{\pi}{2}\right\}$. The sum of all distinct solution of the equation $\sqrt{3}\sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to - **[JEE (Advanced) 2016]**

(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

MTE064

4. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is _____

[JEE (Advanced) 2018]

MTE065

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Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $f(x) = \sin(\pi\cos x)$ and $g(x) = \cos(2\pi\sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$.

List-I contains the sets X, Y, Z and W. List -II contains some information regarding these sets.

List-I		List-II	
(I)	X	(P)	$\supseteq\left\{\frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi\right\}$
(II)	Y	(Q)	an arithmetic progression
(III)	Z	(R)	NOT an arithmetic progression
(IV)	W	(S)	$\supseteq\left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$
		(T)	$\supseteq\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$
		(U)	$\supseteq\left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

5. Which of the following is the only CORRECT combination ? **[JEE (Advanced) 2019]**

(A) (II), (R), (S) (B) (I), (P), (R) (C) (II), (Q), (T) (D) (I), (Q), (U)

MTE066

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6. Which of the following is the only CORRECT combination? **[JEE (Advanced) 2019]**

(A) (IV), (Q), (T) (B) (IV), (P), (R), (S) (C) (III), (R), (U) (D) (III), (P), (Q), (U)

MTE067

7. Let $f:[0,2] \to \mathbb{R}$ be the function defined by

$f(x) = (3 - \sin(2\pi x))\sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$ **[JEE (Advanced) 2020]**

If $\alpha, \beta \in [0,2]$ are such that $\{x \in [0,2]: f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is ____

MTE068

8. Consider the following lists :

	List-I		**List-II**
(I)	$\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P)	has two elements
(II)	$\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(Q)	has three elements
(III)	$\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R)	has four elements
(IV)	$\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S)	has five elements
		(T)	has six elements

The correct option is : **[JEE (Advanced) 2022]**

(A) (I) $\to$ (P); (II) $\to$ (S); (III) $\to$ (P); (IV) $\to$ (S)

(B) (I) $\to$ (P); (II) $\to$ (P); (III) $\to$ (T); (IV) $\to$ (R)

(C) (I) $\to$ (Q); (II) $\to$ (P); (III) $\to$ (T); (IV) $\to$ (S)

(D) (I) $\to$ (Q); (II) $\to$ (S); (III) $\to$ (P); (IV) $\to$ (R)

MTE069

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	C	C	B	A	B	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	D	B	A	A	C	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	B,C	A,B,C,D	A,B,C	B,C	B,D	A,C	B,C	A,C,D	C
Que.	11	12	13	14	15					
Ans.	C	A	B	A	A → S; B → P; C → R; D → Q					

EXERCISE (S)

1.	3	**2.**	3	**3.**	24	**4.**	07	**5.**	17
6.	2	**7.**	5025	**8.**	50	**9.**	6	**10.**	5

EXERCISE - JEE (Main) PYQ

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Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	4	1	1	1	2	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	3	1	3	25	4	1	2
Que.	21	22	23	24	25	26				
Ans.	2	3	4	4	2	3				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8
Ans.	D	8	C	0.50	C	B	1.00	B

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